

24 Multiple Reactions with Heat Effects

Recall the energy balances that we've been looking at so far, where we have:

- Heat Generated term (Q_g or $G(T)$) -- due to the reaction
- Heat Removed term (Q_r or $R(T)$) -- due to the heat exchanger and also sensible heat changes as reactor contents "absorb" heat given off by the reaction

The Heat Generated terms look like: $(r_A)(\Delta H^{rxn})$ energy/mol A ← must be in units of

For the case of multiple reactions, the Heat Generated term looks like: $\sum_i (-r_{ij})(-\Delta H_{ij}^{rxn})$, where i are reactions and j are species.

In this sum, there should be **one $(-r_j)(-\Delta H_j^{rxn})$ term per reaction**. The purpose of the j subscript is to ensure consistent stoichiometry/units ... that each rate of reaction and heat of reaction pair are taken with respect to the same species. What matters here is that j is consistent for each pair; it is OK if j varies from pair to pair.

Beyond that, the solution procedure is analogous to the one you've already used to solve reactor design problems when there are multiple reactions: You have to use N 's or F 's instead of X 's, and because of this you need to use mathematical software.

Let's do an example:

The gas phase elementary reactions $2A \rightleftharpoons B$ and $2B + A \rightarrow C$ are carried out in a non-isothermal PFR with no pressure drop. Write the symbolic expression for the heat generated term.

Info:

$k_{1A} = 50 \text{ M}^{-1} \text{ min}^{-1}$ at 350 K with $E = 8 \text{ kJ/mol}$
 $k_{2C} = 4000 \text{ M}^{-2} \text{ min}^{-1}$ at 310 K with $E = 4 \text{ kJ/mol}$
 $K_{C,A} = 10 \text{ M}^{-1}$ at 315 K
 $\Delta H_{1A}^{rxn} = -20 \text{ kJ/mol A}$
 $\Delta H_{2B}^{rxn} = +30 \text{ kJ/mol B}$

rate of cons if reactant
 $\frac{1}{2}$ rate of gen if product

$$\frac{dT}{dV} = \frac{Q_g + Q_r}{\sum F_i C_{p,i}}$$

$$\left(\frac{\text{moles A}}{\text{L} \cdot \text{min}} \right) \left(\frac{\text{kJ}}{\text{mole A}} \right) = \frac{\text{kJ}}{\text{L} \cdot \text{min}}$$

$$Q_g = \sum_i (-r_{ij}) (-\Delta H_{ij}^{rxn}) = (-r_{1A}) (-\Delta H_{1A}^{rxn})$$

$$+ \left(r_{2C} \right) \left(-\Delta H_{2B}^{rxn} \right) \left(\frac{2 \text{ mol B}}{1 \text{ mol C}} \right)$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $i \quad j \quad i \quad j$

$2(-\Delta H_{2B}^{rxn}) = -\Delta H_{2C}^{rxn}$

$\uparrow \quad \uparrow$
 $i \quad j$

$-20 \frac{\text{kJ}}{\text{mole A}}$
 $+20 \frac{\text{kJ}}{\text{mole A}}$

$$\frac{r_{2C}}{1} = \frac{r_{2B}}{-2}$$

$$r_{2C} = -\frac{1}{2} r_{2B}$$

$$\rightarrow \left(-\frac{1}{2} r_{2B} \right) \left(-\Delta H_{2B}^{rxn} \right)$$

$\uparrow \quad \uparrow$
 $i \quad j$

$+30 \frac{\text{kJ}}{\text{mole B}}$
 $-30 \frac{\text{kJ}}{\text{mole B}}$

$$\frac{1}{2} k_{2C} C_B^2 C_A$$

$\uparrow$

use Arrhenius expression to determine as a function of T.

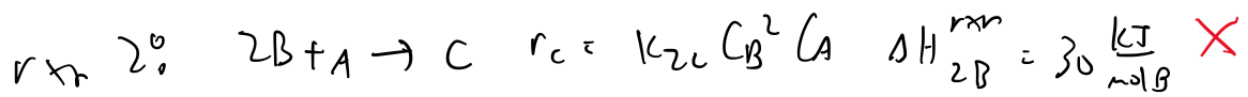
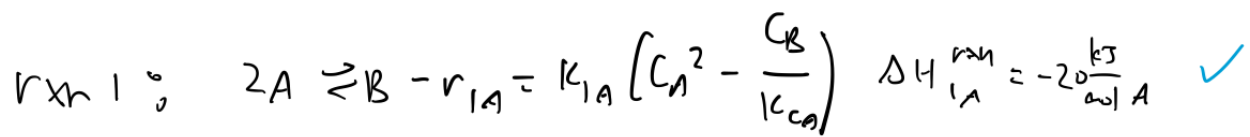
use stoichiometry expressions for concentration

$$\underline{C_j = C_{T0} \left(\frac{F_T}{F_{T0}} \right) \left(\frac{T_0}{T} \right)}$$

$$F_T = F_A + F_B + F_C$$

using matrices

① make sure units on ΔH^{rxn} are consistent
w/ units on rate.



$$\Delta H_{2C}^{rxn} = \left(30 \frac{kJ}{mol B} \right) \left(\frac{2 mol B}{mol C} \right) = 60 \frac{kJ}{mol C}$$

② break reversible rxns up:

new rxn 1: $2A \rightarrow B$

$$-r_{1A} = k_{1A} C_A^2$$

new rxn 2: $B \rightarrow 2A$

$$r_{2A} = \frac{k_{1A}}{k_{2A}} C_B$$

new rxn 3: $2B + A \rightarrow C$

$$r_{3C} = k_{3C} C_B^2 C_A$$

③ create array of ΔH° rxns

$$H = \begin{bmatrix} -20 \\ 20 \\ 60 \end{bmatrix}$$

④ Solve for Q_y°

$$Q_g = r(-H^T)$$

$$r = \begin{bmatrix} K_{1A} C_A^2 \\ \frac{K_{1A}}{K_{CA}} C_B \\ K_{3C} C_B^2 C_A \end{bmatrix}$$

$$r(-H^T) = \begin{bmatrix} K_{1A} C_A^2 \\ \frac{K_{1A}}{K_{CA}} C_B \\ K_{3C} C_B^2 C_A \end{bmatrix} \begin{bmatrix} 20 & -20 & -60 \end{bmatrix}$$

$$Q_g = \left(20 \frac{\text{kJ}}{\text{mol A}} \right) (K_{1A} C_A^2) - \left(20 \frac{\text{kJ}}{\text{mol A}} \right) \left(\frac{K_{1A}}{K_{CA}} C_B \right) - \left(60 \frac{\text{kJ}}{\text{mol C}} \right) K_{3C} C_B^2 C_A$$